

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT: CYBER & INFORMATION SECURITY
PRACTICAL - 3

AIM: Message Authentication Codes:

Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

Code:

```
import hmac
import hashlib

# Function to generate MAC
def generate_mac(key, message):
    return hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()

# Function to verify MAC
def verify_mac(key, message, received_mac):
    calculated_mac = generate_mac(key, message)
    return hmac.compare_digest(calculated_mac, received_mac)

WwQ # User input
key = input("Enter the secret key: ")
message = input("Enter the message: ")

# Generate MAC
mac = generate_mac(key, message)
print("\nGenerated MAC:", mac)

# Verification
print("\n--- Verification ---")
verify_message = input("Enter the message to verify: ")
received_mac = input("Enter the MAC to verify: ")

if verify_mac(key, verify_message, received_mac):
    print("Message Verified Successfully")
else:
    print("Message Verification Failed")
```

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OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/info/OneDrive/Desktop/prac3_cYBER.py =====
Enter the secret key: key123
Enter the message: Pokemon

Generated MAC: 321e2fa6de9ec44e2dc49fc75899534ef6f13aeeacdbf562b0f61760c7341b3a

--- Verification ---
Enter the message to verify: Pokemon
Enter the MAC to verify: 321e2fa6de9ec44e2dc49fc75899534ef6f13aeeacdbf562b0f61760c7341b3a
Message Verified Successfully
>>>
```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\info>C:\Users\info\OneDrive\Desktop\prac3_cYBER.py
Enter the secret key: 3
Enter the message: Sumeet

Generated MAC: f4532d1df737a4aeb7c842baaa48a34a813b50513f3e78502e566026cc46459

--- Verification ---
Enter the message to verify: Sumeet
Enter the MAC to verify: f4532d1df737a4aeb7c842baaa48a34a813b50513f3e78502e566026cc46459
Message Verified Successfully

C:\Users\info>
```

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GUI:

```
import tkinter as tk
from tkinter import ttk, messagebox
import hmac
import hashlib
```

#Functions

```
def generate_mac(key, message):
    return hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()
```

```
def generate():
    key = key_entry.get()
    message = message_entry.get()

    if not key or not message:
        messagebox.showerror("Error", "Please enter the Secret Key and Message.")
        return
```

```
    mac = generate_mac(key, message)
```

```
    generated_mac.config(state="normal")
    generated_mac.delete(0, tk.END)
    generated_mac.insert(0, mac)
    generated_mac.config(state="readonly")
```

```
def verify():
    key = key_entry.get()
    message = verify_message.get()
    mac = verify_mac.get()
```

```
    if not key or not message or not mac:
        messagebox.showerror("Error", "Please fill all verification fields.")
        return
```

```
    calculated = generate_mac(key, message)
```

```
    if hmac.compare_digest(calculated, mac):
        messagebox.showinfo("Verification", "Message Verified Successfully")
    else:
```

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messagebox.showerror("Verification", " Message Verification Failed")

```
#Window
root = tk.Tk()
root.title("HMAC Message Authentication")
root.geometry("720x480")
root.configure(bg="#EAF4FC")
root.resizable(False, False)

# Style
style = ttk.Style()
style.theme_use("clam")

style.configure("TLabel",
                background="#EAF4FC",
                font=("Segoe UI", 11))

style.configure("Title.TLabel",
                font=("Segoe UI", 18, "bold"),
                foreground="#1E3A8A",
                background="#EAF4FC")

style.configure("TButton",
                font=("Segoe UI", 10, "bold"),
                padding=6)

#Title
title = ttk.Label(root,
                  text=" HMAC Message Authentication",
                  style="Title.TLabel")
title.pack(pady=15)

#Generate Frame
gen_frame = tk.LabelFrame(root,
                          text=" Generate MAC ",
                          font=("Segoe UI", 11, "bold"),
                          bg="white",
                          fg="#1E3A8A",
                          padx=15,
                          pady=10)

gen_frame.pack(fill="x", padx=20, pady=10)

ttk.Label(gen_frame, text="Secret Key").grid(row=0, column=0, sticky="w", pady=8)
```

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```
key_entry = ttk.Entry(gen_frame, width=55)
key_entry.grid(row=0, column=1, padx=10)

ttk.Label(gen_frame, text="Message").grid(row=1, column=0, sticky="w", pady=8)

message_entry = ttk.Entry(gen_frame, width=55)
message_entry.grid(row=1, column=1, padx=10)

ttk.Button(gen_frame,
            text="Generate MAC",
            command=generate).grid(row=2, column=1, pady=10)

ttk.Label(gen_frame, text="Generated MAC").grid(row=3, column=0, sticky="w",
pady=8)

generated_mac = ttk.Entry(gen_frame, width=75)
generated_mac.grid(row=3, column=1, padx=10)
generated_mac.config(state="readonly")

#Verification Frame
verify_frame = tk.LabelFrame(root,
                             text=" Verify MAC ",
                             font=("Segoe UI", 11, "bold"),
                             bg="white",
                             fg="#1E3A8A",
                             padx=15,
                             pady=10)

verify_frame.pack(fill="x", padx=20, pady=10)

ttk.Label(verify_frame, text="Message").grid(row=0, column=0, sticky="w", pady=8)

verify_message = ttk.Entry(verify_frame, width=55)
verify_message.grid(row=0, column=1, padx=10)

ttk.Label(verify_frame, text="MAC").grid(row=1, column=0, sticky="w", pady=8)

verify_mac = ttk.Entry(verify_frame, width=75)
verify_mac.grid(row=1, column=1, padx=10)

ttk.Button(verify_frame,
            text="Verify MAC",
            command=verify).grid(row=2, column=1, pady=15)

#Footer
footer = tk.Label(root,
```

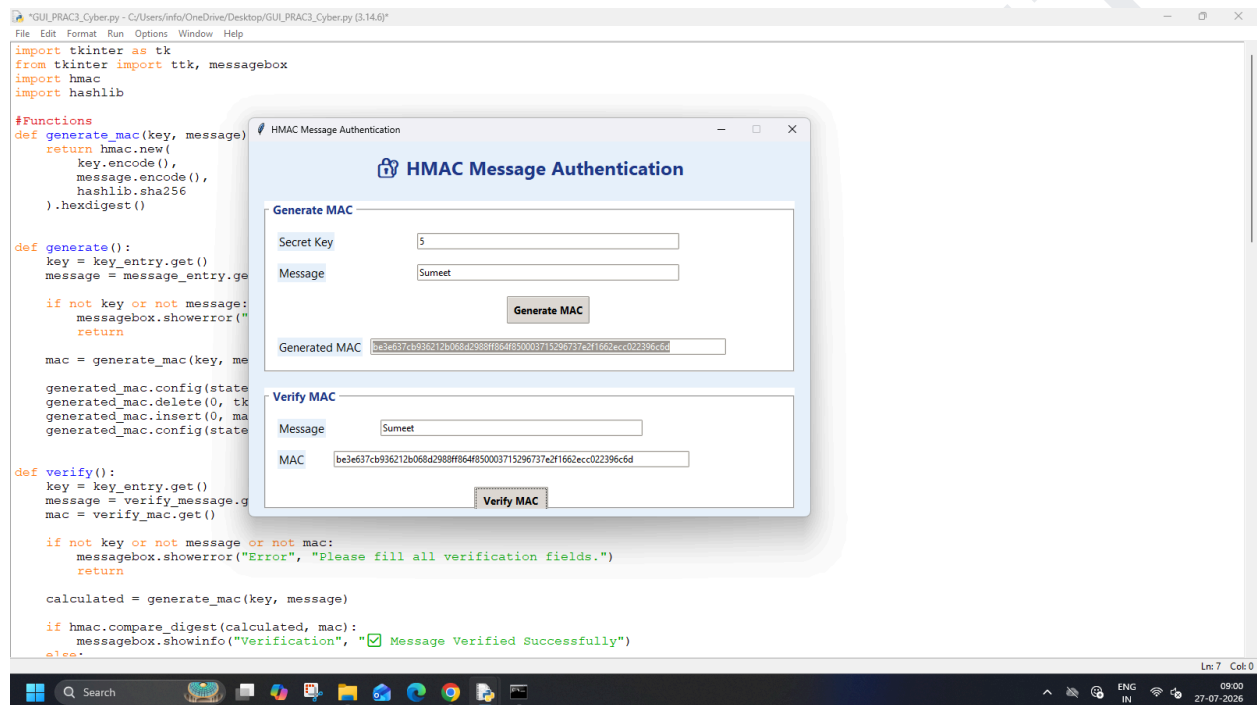
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```
text="HMAC using SHA-256",  
bg="#EAF4FC",  
fg="gray40",  
font=("Segoe UI", 9))
```

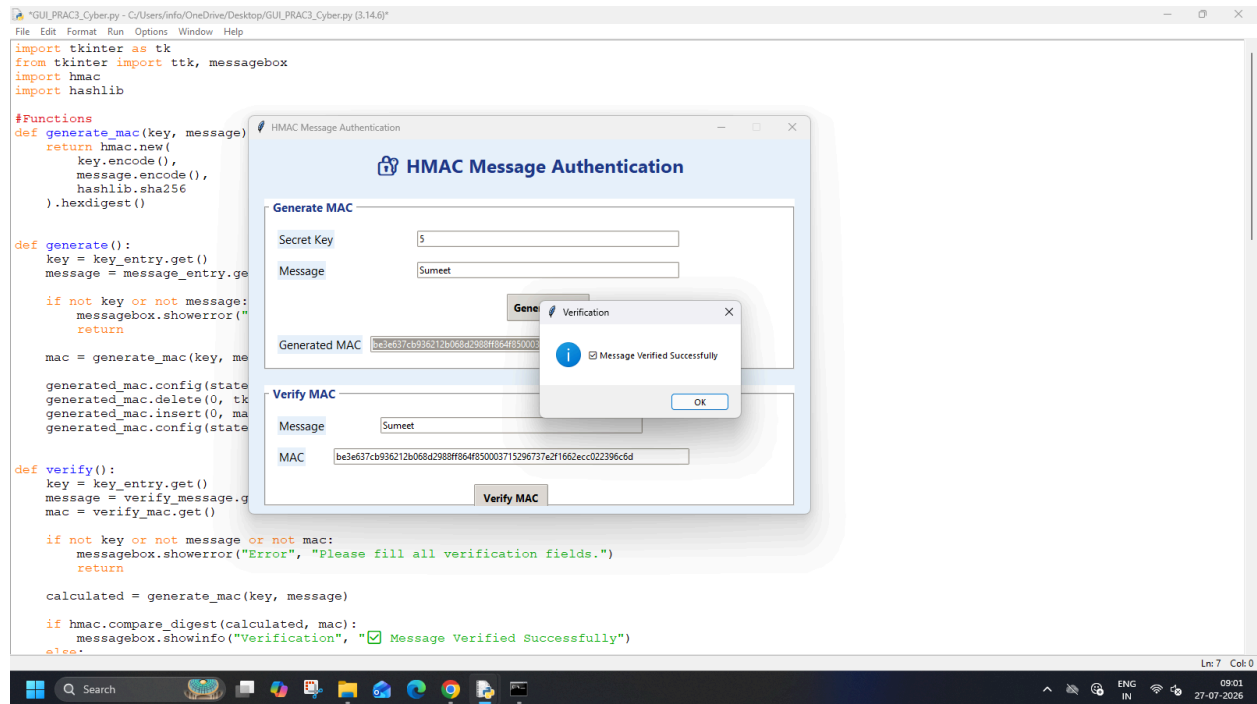
```
footer.pack(side="bottom", pady=10)
```

```
root.mainloop()
```

OUTPUT:



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